

fong-spivak-invitation: residuals

equations measured 2026-09-05T11:48:20Z against the model of 2026-09-03

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_ E00202 [conf 0.001]	291	■ 0.001 ● C +70	G_{1}=\begin{array}{ c } \hline \text{ { 李 } } \\ \hline f=g \\ \end{array} \quad G_{2}=\begin{array}{ c } \hline f \\ \stackrel{\mathcal{Q}}{\bullet} \\ \hline f=a \\ \end{array} \quad G_{3}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ g\downarrow & & \downarrow h \\ \bullet & \xrightarrow{i} & \bullet \\ f\circ h=g\circ i \\ \end{array} \quad G_{4}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ g\downarrow & & \downarrow h \\ \bullet & & \bullet \\ \text{no equations} \\ \end{array}	(not rendered)	
fong-spivak-invitation_ F02349	228	■ 0.185	\operatorname{cospan} \underline{a} \xrightarrow{\operatorname{id}_{\underline{a}}} \operatorname{id}_{\underline{a}} \stackrel{\operatorname{id}_{\underline{a}}}{\longleftarrow} \underline{a}	(not rendered)	(iv) identities $\operatorname{id}_a \in \mathbf{Set}(a; a)$ just given by the identity $\operatorname{cospan} \underline{a} \xrightarrow{\operatorname{id}_{\underline{a}}} \underline{a} \xleftarrow{\operatorname{id}_{\underline{a}}} \underline{a}$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_E00095	146	■ 0.151 ● C +37	$\operatorname{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X} \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a,b) & \text{if } a,b \in \mathcal{Y} \end{cases}$	$\operatorname{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X} \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a,b) & \text{if } a,b \in \mathcal{Y} \end{cases}$	$\operatorname{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X}, \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ y(a,b) & \text{if } a,b \in \mathcal{Y}. \end{cases}$
fong-spivak-invitation_E00134	213	■ 0.235 ● C +91	$\begin{array}{l} \operatorname{in}(\mu) := 2, \\ \& \operatorname{in}(\eta) := 0, \operatorname{in}(\delta) := 1, \\ \& \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, \\ \operatorname{out}(\eta) := 1, \\ \operatorname{out}(\delta) := 2, \\ \operatorname{out}(\epsilon) := 0. \end{array}$	$\operatorname{in}(\mu) := 2, \operatorname{in}(\eta) := 0, \operatorname{in}(\delta) := 1, \operatorname{in}(\epsilon) := 1,$	$\begin{array}{llll} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$
fong-spivak-invitation_E00227	306	■ 0.259 ● C +43	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x,a) \otimes \mathcal{X}(a,b) \otimes \mathcal{X}(c,d) \otimes \mathcal{X}(b,c) \otimes \mathcal{X}(d,e) \otimes \mathcal{X}(e,y) \\ & = \mathcal{X}(x,y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a,1),(b,c,d)) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} x(x,a) \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ & = x(x,y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a,1),(b,c,d)) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x,a) \otimes \mathcal{X}(a,b) \otimes \mathcal{X}(c,d) \otimes \mathcal{X}(b,c) \otimes \mathcal{X}(d,e) \otimes \mathcal{X}(e,y) \\ & = \mathcal{X}(x,y), \end{aligned}$
fong-spivak-invitation_E00075 (3.15)	121	■ 0.380 ● C +40	$\mathcal{G} := \mid \stackrel{\text{Email}}{\text{received_by}} \stackrel{\text{sent_by}}{\text{Address}}$	<i>(not rendered)</i>	$\mathcal{G} := \begin{array}{c} \text{Email} \xrightarrow{\text{sent_by}} \text{Address} \\ \text{received_by} \\ \text{no equations} \end{array}$
fong-spivak-invitation_E00010 (1.7)	44	■ 0.623 ● C +21	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-invitation_E00143 (6.21)	226	■ 0.754 ● C +20	$\begin{aligned} & \circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \\ & \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t); \end{aligned}$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \\ \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \\ \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$
fong-spivak-invitation_E00131	206	■ 0.985 ● C +28	$\operatorname{colim}_{\mathcal{J}} D := \{(v,d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v,d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v,d) \mid v \in V \text{ and } d \in D(v)\} / \sim$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

69 more flagged rows below the band, stated as a count: 40 component, 29 weak; confidence high 37, mid 12, low 20, none 0

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation-E00135 [conf 0.019]	215	■ 0.019 ●	\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\mathrm{id}_X} \mathrm{id}_X X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xrightarrow{\mathrm{id}_X} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}	$\mu_X := \left(X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X \right)$ $\eta_X := \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right)$ $\delta_X := \left(X \xrightarrow{\mathrm{id}_X} X \xrightarrow[\mathrm{id}_X, \mathrm{id}_X]{} X + X \right)$ $\epsilon_X := \left(X \xrightarrow{\mathrm{id}_X} \emptyset \right)$	$\begin{aligned} \mu_X &:= (X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X) \\ \eta_X &:= (\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X) \\ \delta_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X) \\ \epsilon_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset) \end{aligned}$
fong-spivak-invitation-E00199 [conf 0.051]	291	■ 0.051 ● S +0	A, \quad A \text{ : } f, \quad A \text{ : } g, \quad A \text{ : } f : h, \quad A \text{ : } g : i, \quad B, \quad B : h, \quad C, \quad C : i, \quad D	$A, \quad A : f, \quad A : g, \quad A : f : h, \quad A : g : i, \quad B, \quad B : h, \quad C, \quad C : i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ f \circ h, \quad A \circ g \circ i, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation-E00257 (A.2) [conf 0.064]	323	■ 0.064 ● S -1	\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{\mathrm{id}}{\longleftarrow} X + X \xrightarrow{\mathrm{id}} X \stackrel{\mathrm{id}}{\longleftarrow} X	$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longleftarrow} X + X \xrightarrow{[\mathrm{id}, \mathrm{id}]} X \stackrel{\mathrm{id}}{\longleftarrow} X$	$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longleftarrow} X + X \xrightarrow{[\mathrm{id}, \mathrm{id}]} X \stackrel{\mathrm{id}}{\longleftarrow} X$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					